

Transshipment: optimality conditions

Michel Bierlaire

Solution of the practice quiz

1. A dual variable λ_i is associated with each node of the transformed network, and a dual variable μ_{ij} is associated with each arc of the transformed network. The Lagrangian is

$$L(x, \lambda, \mu) = \sum_{(i,j) \in \mathcal{A}} c_{ij}x_{ij} + \sum_i \lambda_i \left(\sum_{(i,j) \in \mathcal{A}} x_{ij} - \sum_{(k,i) \in \mathcal{A}} x_{ki} - s_i \right) - \sum_{(i,j) \in \mathcal{A}} \mu_{ij}x_{ij},$$

that is

$$\begin{aligned} L(x, \lambda, \mu) = & 100x_{2,B} + 50x_{1,NY} + 80x_{3,B} \\ & + \lambda_{LA}(-x_{1,LA} - x_{2,LA} + 30) + \lambda_1(x_{1,NY} + x_{1,LA} - 40) \\ & + \lambda_{NY}(-x_{3,NY} - x_{1,NY} + 40) + \lambda_2(x_{2,LA} + x_{2,B} - 40) \\ & + \lambda_B(-x_{2,B} - x_{3,B} + 50) + \lambda_3(x_{3,NY} + x_{3,B} - 40) \\ & - \mu_{1,NY}x_{1,NY} - \mu_{1,LA}x_{1,LA} - \mu_{2,LA}x_{2,LA} \\ & - \mu_{2,B}x_{2,B} - \mu_{3,B}x_{3,B} - \mu_{3,NY}x_{3,NY} \\ = & x_{1,LA}(-\lambda_{LA} + \lambda_1 - \mu_{1,LA}) + x_{1,NY}(50 - \lambda_{NY} + \lambda_1 - \mu_{1,NY}) \\ & + x_{2,LA}(-\lambda_{LA} + \lambda_2 - \mu_{2,LA}) + x_{2,B}(100 - \lambda_B + \lambda_2 - \mu_{2,B}) \\ & + x_{3,NY}(-\lambda_{NY} + \lambda_3 - \mu_{3,NY}) + x_{3,B}(80 - \lambda_B + \lambda_3 - \mu_{3,B}) \\ & + 30\lambda_{LA} + 40\lambda_{NY} + 50\lambda_B - 40\lambda_1 - 40\lambda_2 - 40\lambda_3. \end{aligned}$$

The coefficient of each flow variable must be zero to get a bounded dual

problem:

$$\begin{aligned}
-\lambda_{LA} + \lambda_1 &= \mu_{1,LA}, \\
50 - \lambda_{NY} + \lambda_1 &= \mu_{1,NY}, \\
-\lambda_{LA} + \lambda_2 &= \mu_{2,LA}, \\
100 - \lambda_B + \lambda_2 &= \mu_{2,B}, \\
-\lambda_{NY} + \lambda_3 &= \mu_{3,NY}, \\
80 - \lambda_B + \lambda_3 &= \mu_{3,B}.
\end{aligned}$$

Moreover since $\mu_{ij} \geq 0, \quad \forall i, j$, we can eliminate the μ 's, and we obtain the following set of constraints:

$$\begin{aligned}
-\lambda_{LA} + \lambda_1 &\geq 0, \\
\lambda_{NY} - \lambda_1 &\leq 50, \\
-\lambda_{LA} + \lambda_2 &\geq 0, \\
\lambda_B - \lambda_2 &\leq 100, \\
-\lambda_{NY} + \lambda_3 &\geq 0, \\
\lambda_B - \lambda_3 &\leq 80.
\end{aligned}$$

The objective function is:

$$q(\lambda) = 30\lambda_{LA} + 40\lambda_{NY} + 50\lambda_B - 40\lambda_1 - 40\lambda_2 - 40\lambda_3.$$

The dual problem is therefore:

$$\max 30\lambda_{LA} + 40\lambda_{NY} + 50\lambda_B - 40\lambda_1 - 40\lambda_2 - 40\lambda_3$$

subject to

$$\begin{aligned}
-\lambda_{LA} + \lambda_1 &\geq 0, \\
\lambda_{NY} - \lambda_1 &\leq 50, \\
-\lambda_{LA} + \lambda_2 &\geq 0, \\
\lambda_B - \lambda_2 &\leq 100, \\
-\lambda_{NY} + \lambda_3 &\geq 0, \\
\lambda_B - \lambda_3 &\leq 80.
\end{aligned}$$

2. The optimality conditions are

- $c_{ij} + \lambda_i - \lambda_j \geq 0, \quad \forall (i, j) \in \mathcal{A}.$
- $c_{ij} + \lambda_i - \lambda_j = 0, \quad \forall (i, j) \in \mathcal{A} \text{ such that } x_{ij} > 0.$

We are giving tentative values for λ in such a way that they are consistent with the optimality conditions.

First, we remember that one λ can be normalized to an arbitrary value. We set

$$\lambda_1 = 0.$$

We consider the two arcs leaving node 1. If we set the dual variable of the downstream node to

$$\lambda_j = \lambda_i + c_{ij},$$

the conditions will be verified. Therefore,

$$\begin{aligned}\lambda_{LA} &= \lambda_1 + c_{1,LA} = 0 + 0 = 0, \\ \lambda_{NY} &= \lambda_1 + c_{1,NY} = 0 + 50 = 50.\end{aligned}$$

Consider now the arc (3, NY). If we set the dual variable of the upstream node to

$$\lambda_i = \lambda_j - c_{ij},$$

the conditions will be verified. Therefore,

$$\lambda_3 = \lambda_{NY} - c_{3,NY} = 50 - 0 = 50.$$

We consider now the arc (3, B) and we obtain

$$\lambda_B = \lambda_3 + c_{3,B} = 50 + 80 = 130.$$

For arc (2, B), we have

$$\lambda_2 = \lambda_B - c_{2,B} = 130 - 100 = 30.$$

Finally, for arc (2, LA), we have

$$\lambda_{LA} = \lambda_2 + c_{2,LA} = 30 + 0 = 30.$$

But earlier, we had decided to set λ_{LA} to 0. The inequality must be verified for both arcs. Therefore, we must have:

$$\begin{aligned}\lambda_{LA} &\leq \lambda_1 + c_{1,LA} = 0, \\ \lambda_{LA} &\leq \lambda_2 + c_{2,LA} = 30.\end{aligned}$$

Consequently, the inequalities will be verified is $\lambda_{LA} = 0$, and the dual variables are

$$\begin{aligned}\lambda_1 &= 0, \\ \lambda_2 &= 30, \\ \lambda_3 &= 50, \\ \lambda_{LA} &= 0, \\ \lambda_{NY} &= 50, \\ \lambda_B &= 130.\end{aligned}$$

The optimality conditions are summarized in the following table:

Arc	$c_{ij} + \lambda_i - \lambda_j$
(1, LA)	$0 + 0 - 0 = 0$,
(1, NY)	$50 + 0 - 50 = 0$,
(2, LA)	$0 + 30 - 0 = 30$,
(2, B)	$100 + 30 - 130 = 0$,
(3, B)	$80 + 50 - 130 = 0$,
(3, NY)	$0 + 50 - 50 = 0$.

We note that the arc (2, LA) is the only one where the inequality is strict. Therefore, we now that this arc carries no flow:

$$x_{2,LA}^* = 0.$$

Therefore, in order to verify the flow conservation constraints at node LA, the demand must be served by the arc (1, LA), so that

$$x_{1,LA}^* = 30.$$

At node 1, the flow conservation constraints are verified if

$$x_{1,NY}^* = 10.$$

Applying the flow conservation constraints at node NY, we obtain

$$x_{3, \text{NY}}^* = 30.$$

At node 3, we obtain

$$x_{3, B}^* = 10.$$

At node B, we obtain

$$x_{2, B}^* = 40.$$

And we can check that the flow conservation constraints are verified at node 2. If we now ignore the slack variables, and come back to the original formulation, this can be interpreted as:

- sending 10 units of flow from LA to NY, for a total cost of 500,
- sending 40 units of flow from LA to B, for a total cost of 4000,
- sending 10 units of flow from NY to B, for a total cost of 800.

The total cost is therefore 5300. Note that the objective function of the dual is

$$\begin{aligned} q(\lambda) &= 30\lambda_{\text{LA}} + 40\lambda_{\text{NY}} + 50\lambda_B - 40\lambda_1 - 40\lambda_2 - 40\lambda_3, \\ &= 0 + 2000 + 6500 + 0 - 1200 - 2000 \\ &= 5300, \end{aligned}$$

consistently with the strong duality theorem.